UNIT 5 · 9 hours

Op-Amp Specifications



Op-Amp and Linear IC Applications

Basic Electrical and Electronics Engineering (EEE103)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
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Learning objectives

- Define key op-amp specifications
- Interpret typical values for the 741
- Calculate slew-rate limits
- Understand offset voltage and bias currents

Syllabus context

Introduction, op-amp specifications, differential and common-mode operation, op-amp applications: inverting/non-inverting amplifier, summing, integrator, differentiator, comparator, 555 timer IC in astable mode, 78XX series IC voltage regulators.

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Important specifications

Parameter	Definition	741 typical
Input offset voltage	Differential input needed to make output zero	2 mV
Input bias current	Average of the two input currents	80 nA
Input offset current	Difference of the two input currents	20 nA
CMRR	Ad / Acm	90 dB
PSRR	ΔVsupply / ΔVout (input referred)	30 µV/V
Slew rate	Maximum dVo/dt	0.5 V/µs
Output swing	Max undistorted peak output	±13 V

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Input-related specifications

Input impedance

Ratio of input voltage change to input current change at one terminal. Very high (MΩ–TΩ).

Input offset voltage

Ideally both inputs grounded → vo = 0. In practice a small offset appears; a differential voltage must be applied to null it.

Input bias and offset currents

Small DC currents flowing into the input transistors; cause output errors with high-value resistors.

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Output and frequency specifications

Output impedance

Ratio of output voltage change to output current change; low (≈ 75 Ω for the 741).

Output voltage swing

Peak output without distortion; slightly less than the supply: ±13 V on ±15 V.

Bandwidth and slew rate

Gain-bandwidth product ≈ 1 MHz; slew rate = speed limit of the output.

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Slew rate and CMRR formulas

$SR = (dVo/dt)max$

V/μs

$f_{max} = SR / (2\pi V_p)$

full-power bandwidth

$CMRR = A_d / A_{cm}$

in dB: 20 log10(Ad/Acm)

- For a sine of peak Vp, $(dVo/dt)max = 2\pi f V_p$.

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Numerical: slew-rate limit

Problem A 741 (SR = 0.5 V/μs) amplifies a sine of peak output 10 V. What is the highest undistorted frequency?

- $f_{max} = SR / (2\pi V_p)$
- $= 0.5 \times 10^6 \text{ V/s} / (2\pi \times 10 \text{ V})$

Answer: $f_{max} \approx 7.96$ kHz

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Key takeaways

- Offset voltage and bias currents cause output errors
- CMRR = Ad/Acm (often in dB)
- Slew rate limits output speed: fmax = SR/(2πVp)
- 741: ±13 V swing, 1 MHz, 0.5 V/µs

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Practice questions

Q1. Define input offset voltage and bias current.

Q2. Find the maximum frequency for a 5 V peak output with SR = 1 V/µs.

Q3. Define CMRR and slew rate.

Q4. What limits the large-signal bandwidth of an op-amp?